

Supplementary material for Anomalous Diffusion of Deformable Particles in a Honeycomb Network

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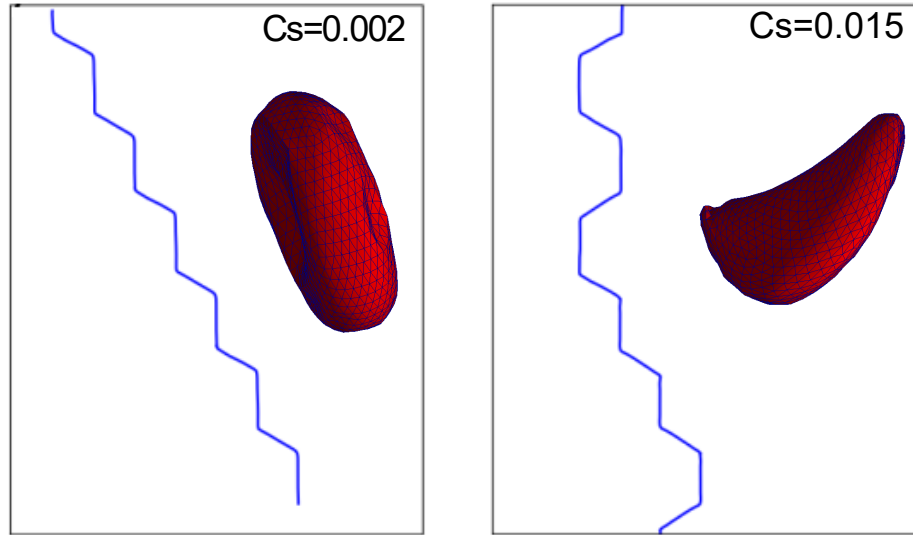


FIG. S1. Trajectories in 3D for different values of Cs . A 3D model of RBC endowed with shear elasticity [1, 2] is used in the simulations. The reduced volume is defined as $v = [V/(4\pi/3)]/[\mathcal{A}/(4\pi)]^{3/2}$. $\mathcal{A}$ and V are the area and volume of the particle. According to the capillary number Ca in 2D model, here we define a capillary number as $Cs = \eta R \dot{\gamma} / \kappa_s$, with κ_s being the shear elastic modulus of RBC cytoskeleton. The particle radius R is defined as $R = [V/(4\pi/3)]^{1/3}$. The channel in the simulations has a square cross section.

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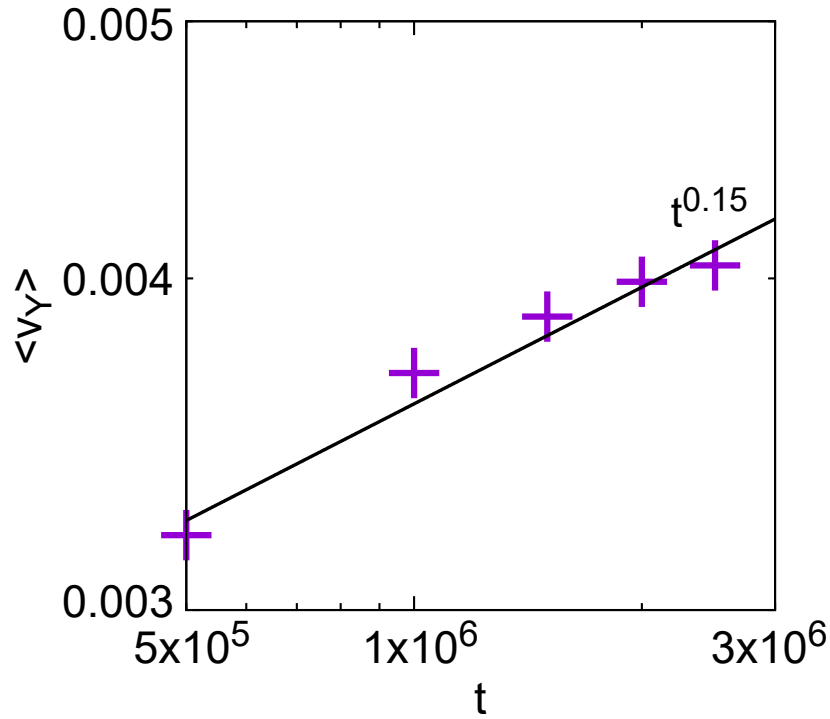


FIG. S2. Time evolution of mean fluid velocity in feeding channels during the particles spreading (for $Ca = 2$). The local concentration in the feeding vessels becomes weaker with time because those vessels feed the lateral spreading of the front at the expense of a reduction of the concentration. Since the driving pressure is fixed, the flow rate increases with time in the feeding vessels due to the local concentration decline. To quantify this effect, we plot the flow rate in the feeding vessels versus time, showing a power law increase $t^{0.15}$. This time-dependent speed naturally amplifies the front speed, explaining the apparent tendency of a super-diffusive and super-ballistic nature shown in Fig. 4 in the main text.

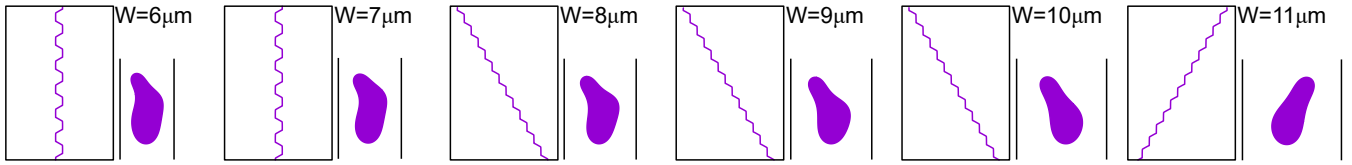


FIG. S3. Trajectories of a single particle in the network with different channel width (for $Ca = 2.5$). The typical shapes of the particle at the middle of a feeding channel are shown. The particle exhibits deterministic lateral migration for a large channel width. When the channel width decreases, the particle inclination tends to be suppressed due to the deformation arising from the confinement. This makes the particle shape more symmetrical to the channel centre line. Below a certain value ($W < 8 \mu m$), the trajectory becomes zigzag similar to the rigid circle situation.

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 - [2] T. Krüger, S. Frijters, F. Günther, B. Kaoui, and J. Harting, Numerical simulations of complex fluid-fluid interface dynamics, *The European Physical Journal Special Topics* **222**, 177 (2013).